

Panyam Lakshmi Narasimhudu

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Professional Summary

Full Stack Developer who builds and ships real products end-to-end, from database to deployed UI. Delivered a real-time AI-powered application (92% core model accuracy, sub-300ms latency) and a full-stack AI travel-planning platform integrating multiple third-party APIs. Proficient in React.js, Node.js, Next.js, Java, Python, MySQL, MongoDB, and REST API design, with a strong foundation in DSA, OOP, system design, Agile workflows, Docker, and clean, production-ready code.

Education

Master of Computer Applications (MCA) Sep 2024 – May 2026
Mohan Babu University – Tirupati

- CGPA: 9.1 / 10.0
- Coursework: Artificial Intelligence, Machine Learning, NLP, Data Structures & Algorithms

Bachelor of Science (Mathematics, Physics, Computer Science) Nov 2021 – Jun 2024
S V Arts College – Tirupati

- CGPA: 7.9 / 10.0
- Coursework: Java, HTML, CSS, Database Management, OOP & MVC Design Principles

Technical Skills

- **Languages:** Java, Python, JavaScript
- **Design & Architecture:** OOP, REST/RESTful API Design, Data Structures & Algorithms, System Design
- **Backend:** Python, Node.js, Next.js, FastAPI
- **Database:** MySQL, MongoDB
- **Frontend:** HTML5, CSS3, React.js
- **Python Libraries:** OpenCV, MediaPipe, NumPy, TensorFlow
- **Tools & Practices:** Git, GitHub, VS Code, Docker, AWS (deployment-ready)

Projects

AI Smart Travel Planner – AI-Powered Travel Planning Web Application Live Demo

- Built a full-stack AI travel planning application using React.js, FastAPI (Python), MongoDB Atlas, and Gemini AI to generate personalized itineraries based on destination, budget, and preferences
- Integrated Google Maps Places & Geocoding APIs and OpenWeather API, and designed REST APIs for authentication, trip management, expense estimation, and recommendations with MongoDB Atlas persistence
- Containerized the application with Docker and prepared it for AWS deployment following scalable, production-ready architecture

AI Software Engineering Assistant – AI-Powered Developer Debugging Assistant GitHub

- Built a local AI-powered developer debugging assistant using Python, FastAPI, React.js, TF-IDF, and Logistic Regression to classify software engineering errors across 10 categories with confidence scoring
- Implemented Retrieval-Augmented Generation (RAG) using ChromaDB and Sentence Transformers (all-MiniLM-L6-v2) to retrieve relevant technical documentation and provide context-aware debugging guidance
- Integrated Ollama with Qwen2.5 0.5B for local AI response generation and implemented persistent chat history using SQLite
- Verified with 17/17 backend tests passed, a passing frontend production build, and an end-to-end ML to RAG to Ollama workflow

VSign – Vision-Based Real-Time Sign Language Recognition and Subtitling System

- Built and trained a real-time gesture recognition system (89 hand gestures) using Python, OpenCV, MediaPipe, and a Multi-Layer Perceptron (MLP) classifier, achieving 92% prediction accuracy
- Engineered a low-latency (200–300 ms) subtitle pipeline with a scalable, accessibility-focused interface

Experience

Python Full Stack Development Intern Jan 2024 – May 2024
Sattva Infotech – Tirupati

- Developed and maintained responsive web pages using HTML5, CSS3, and JavaScript following MVC-aligned design principles
- Built and integrated backend components in Python, contributing to functional, scalable feature delivery
- Collaborated with senior developers in an Agile team setting, participating in code reviews to strengthen code quality

Certifications

Web Development and UI/UX Fundamentals Certificate View Certificate